Building Classification From Aerial Imagery

Connor Williams
Brigham Young University
Department of Civil and Construction Engineering
Provo, Utah 84604

Gregory Macfarlane
Brigham Young University
Department of Civil and Construction Engineering
Provo, Utah 84604

Kaylen Turley
Brigham Young University
Department of Civil and Construction Engineering
Provo, Utah 84604

***Abstract*—This document is a template demonstrating the Ieee-conference format.**

***Index Terms*—template; demo**

I. Introduction

Travel forecasting models are based heavily on data provided by government statistical agencies, such as the US Census Bureau and the Bureau of Labor Statistics. These agencies provide the land use data needed to construct these models, giving statistics such as household number, median household income, etc. for census areas. In countries where statistical agencies are not as reliable, travel forecasting models are generally not developed or have data gaps. In countries where they are availible, these models are highly reliant on the public release of such statistical data, which leads to problems if such survies are delayed or canceled.

Recent research has sought to find alternative methods to collect land use characterists. Gathering, analysing, and utilizing remote sensing data from satelites has been a major focus over the last several decades. Satelite aerial imagry has been used to anaylize the volume and desnity man-made sturctures and roads to estiamte area populations, in both urban and rural settings[Allaw et al. [1]][2]. Day time imagery has also been used to track vehicle movement between major cities as an indication of economic activity [3]. It has also been used to classify land use by estimating landscape patterns and building fuctions, dividing different surfaces, and then categorizing building types [4]. Night time light data has also been shown to be useful as well. One study coorelated city light output with social economic varibles such as population, GDp and electic power conumption [5], while another used the light emited from cars at night on highways to predict vehicle travel between cities [6]. Other satelite data, such as

Building Footprints [7] - Inventoried residential structures based off microsoft building footprint data (satelite), LiDAR (plane), and point of interest data. When from census level data to annomynized individual house holds.

More recently, with the advancement of more easily accessible deep learning models, machine learning has become very prelivent in analysing and classiflying remote sensing data. It [3] It has even been used to predict population characterists and locations of man-made structures across entire countires. [2]

[8] -Using AI model YOLO, object detection deep learning was used to extract building footprints. [9] - Extracting building footprints from point cloud data. Building were classified before extracting the footprints. Trains an AI model to classify

With the rise of commercially availible and affordable aerial drones, the ability to capture remote sensing data such as high quality aerial imagery has become signifigantly easier.

Developing opensource software for drone use [10] - Opensource hardware and software for drone use.

This leads to the question "can remote sensing data be used to develop land use data to supplement or replace data from statistical agencies?" In this paper the process , and the results of the trained model will be compared to other cities in Utah.

This paper presents a model that recongizes and categorizes building types based off remote sensing data and estimates household statistics for a general area.

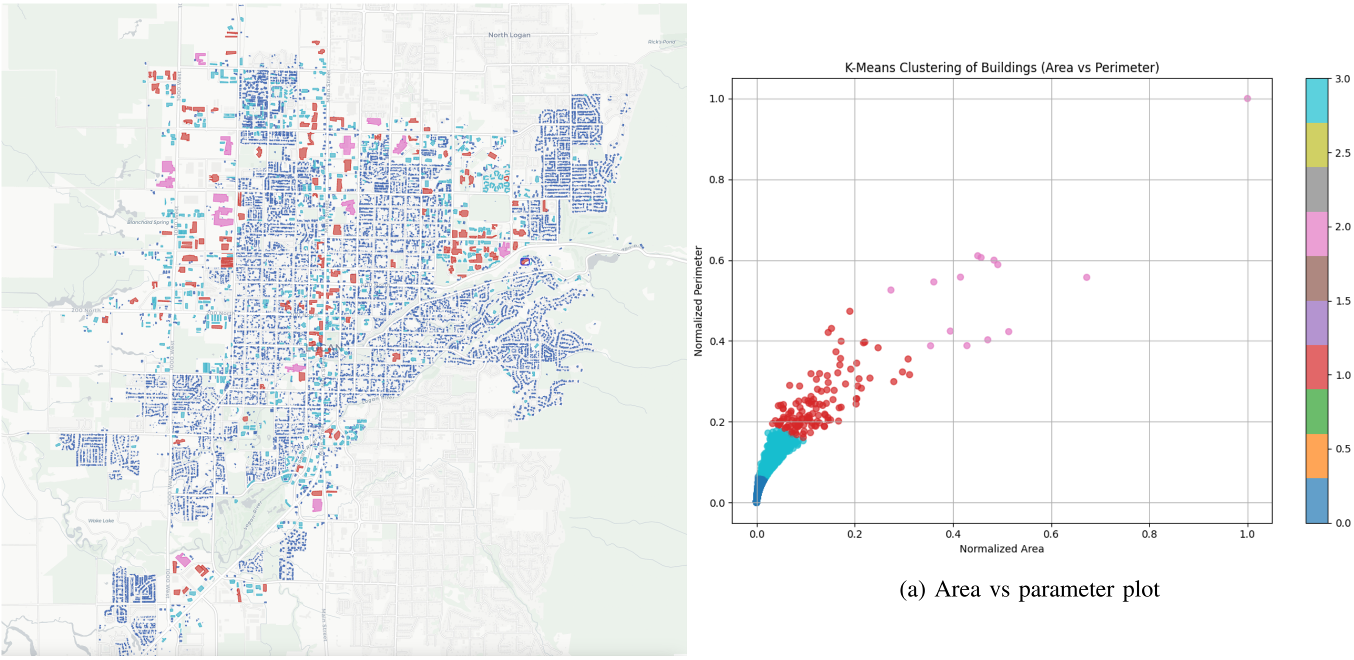
II. Methodolgy

A. Data Collection

The city of Logan, Utah was chosen for analysis. Building footprints for Logan were obtained from the Microsoft Building Footprint public dataset. This dataset allowed for easy access to building footprint data for this inital study.

The latitude and longitude of the city boundries was taken. All building footprints in the city boundary were extracted from the Microsoft Building Footprint dataset and saved as a geojson file. K-mean analysis was run to seperate different sizes of buildings. Optimal K value was found using the elbow method. For Logan the number is five. K-mean analysis was run comparing the building footprint areas to their perimiters. As seen in Figure 1...

Using the building foorprint location and "contextily", 600 randomly chosen aerial images were taken from the internet and croped to focus on a single building. 200 of the images were from clusters that included only larger buildings to help balance building types in the dataset.



(a) Area vs parameter plot

Figure 1: K-mean analysis for Logan, Utah

Using open source tool Label Studio, the images were mannually annotated. Every building in each picture was labeld into five building categories: Appartments, Residential, Commercial, Church, and School.

B. Model Training

Training the Ultraylitics You Only Look Once (YOLO) model based off labeled images

Images were split, 80% used in training and 20% in validation.

Image sets with clusters that included larger building sizes were specifically chosen to make sure the model would be trained on these less common building types.

C. Model Validation

Using trained model to predicit building types in Logan Utah

Choosing cities in Utah using the American Community Survey data. Based off median income level and age of homes.

Using the model to predict building types in other cities in utah based.

III. Results and Discussion

Confusion matrix for Logan - Training and Validation As seen in Figure 2…

The model could identify accurently most building in Logan. It struggled with schools and churches. This is likely due to the very low number of images of both building types provided during training.

Confusion matrix for other analysis cities. What is the results? Was there a difference between how the model did between the cities? Was age of the house a big factor? Was income? Is there something else?

Will it be easier to develop a model to be used at different cities, or a process for people to develop thier own models?

Talk about how this data is still subject to many of the problems the survey data still is. However, much of this research is supseptible to similar probelems as transporation models. Satalitght remote sensing data is not consistent around the world,

1) Subsubsection Heading Here: Subsubsection text here.

IV. Conclusion and Future Research

The conclusion goes here.

As discussed by Hall et al., there are potential ethical problems with such data collection. It strattles the line between public and private information. This will need to be considered as further reserach is done into this area as to stay on the right side of the law.[11]

What does the future research look like?

Training models to detect other things such as cars, swing sets, patios, etc.

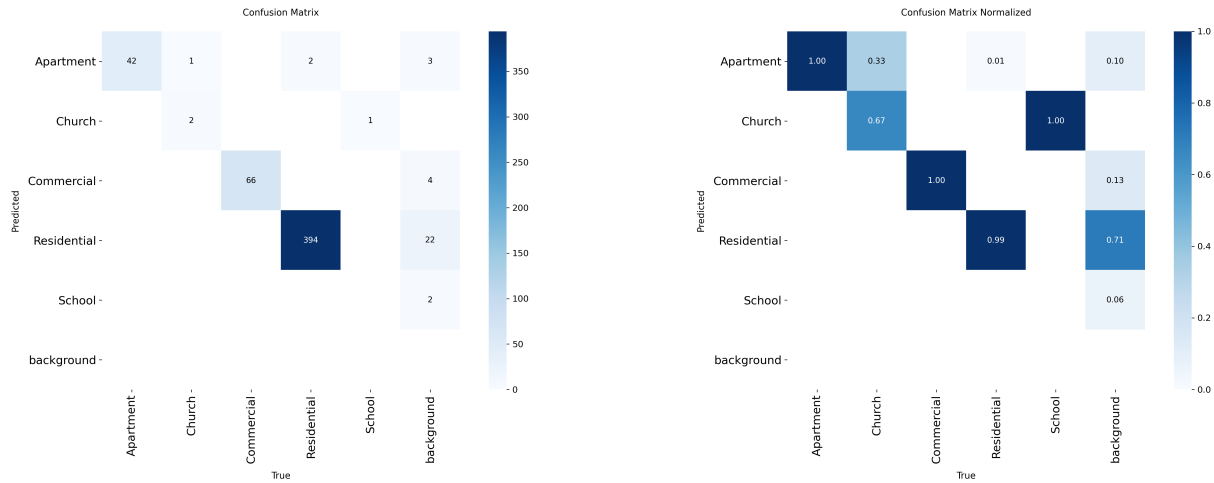
Creating the process to predict household types based off the collected data

Incorperating drone aerial imagery and extracing building footprints from that.

Research into other remote sensing data and how it could be incorperated.

V. Acknowledgment

The authors would like to thank…



(a) Logan confusion matrix

(b) Logan confusion Matrix normalized

Figure 2: YOLO confusion matrixes for Logan, Utah.

References

[1] Kamel Allaw et al. “Population estimation using geographic information system and remote sensing for unorganized areas”. In: *Geoplanning: Journal of Geomatics and Planning* 7.2 (Nov. 1, 2020), pp. 75–86.

[2] Casper Samsø Fibæk, Carsten Keßler, and Jamal Jokar Arsanjani. “A multi-sensor approach for characterising human-made structures by estimating area, volume and population based on sentinel data and deep learning”. In: *International Journal of Applied Earth Observation and Geoinformation* 105 (Dec. 25, 2021), p. 102628.

[3] Patrick Lehnert et al. “Proxying economic activity with daytime satellite imagery: filling data gaps across time and space”. In: *PNAS Nexus* 2.4 (Apr. 1, 2023), pgad099.

[4] Ye Zhang et al. “Landscape patterns and building functions for urban land-use classification from remote sensing images at the block level: A case study of wuchang district, wuhan, china”. In: *Remote Sensing* 12.11 (Jan. 2020), p. 1831.

[5] Shankar Acharya Kamarajugedda, Pradeep V Mandapaka, and Edmond Y.M Lo. “Assessing urban growth dynamics of major southeast asian cities using nighttime light data”. In: *International Journal of Remote Sensing* 38.21 (Nov. 2, 2017), pp. 6073–6093.

[6] Ying Chang et al. “A novel method of evaluating highway traffic prosperity based on nighttime light remote sensing”. In: *Remote Sensing* 12.1 (Jan. 2020), p. 102.

[7] Xinyue Ye et al. “Enhancing population data granularity: A comprehensive approach using lidar, POI, and quadratic programming”. In: *Cities* 152 (Sept. 2024), p. 105223.

[8] Wahidya Nurkarim and Arie Wahyu Wijayanto. “Building footprint extraction and counting on very high-resolution satellite imagery using object detection deep learning framework”. In: *Earth Science Informatics* 16.1 (Mar. 1, 2023), pp. 515–532.

[9] Mayank Sharma and Rahul Dev Garg. “Building footprint extraction from aerial photogrammetric point cloud data using its geometric features”. In: *Journal of Building Engineering* 76 (Oct. 1, 2023), p. 107387.

[10] Gustavo Pereyra Irujo et al. “Open science drone toolkit: open source hardware and software for aerial data capture”. In: *PLOS ONE* 18.4 (Apr. 6, 2023), e0284184.

[11] Ola Hall and Ibrahim Wahab. “The use of drones in the spatial social sciences”. In: *Drones* 5.4 (Dec. 2021), p. 112.